PEng Work to Date: Summary for Approval

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1 Summary

The PEng pipeline now groups detected joints to within 0.0004–0.0010 PGA of HigherHRNet's own grouping, 0.0017–0.0027 of YOLO11x-pose's and 0.0064–0.0066 of RTMO-L's, over two training runs. One trained model serves all three detectors with no retraining. The MEng pipeline trailed all three by about 0.02 PGA. The COCO keypoint AP gap fell from 0.099–0.136 to 0.0024–0.0094, but stays above zero on every detector in both runs.

Two ideas produced this. The first treats grouping as a constrained assignment problem instead of clustering. The second is a hierarchical PG-GAT, trained on detection-like input, with image evidence along candidate bones. Six later improvement attempts were tested and none was adopted, so the pipeline has not changed since. Every number from the final pipeline and its stages was recomputed on the CPU, and the final recipe was trained twice to measure training noise.

Approval is sought to write the full document on this scope.

Terms.

- **PGA** (pose grouping accuracy): the fraction of joints assigned to the correct person, after aligning clusters with annotated people, over detections matched to annotations within 10 px. It is scored on one fixed scene set per detector (2,166 HigherHRNet, 2,196 YOLO11x-pose, 2,121 RTMO-L), so every row of a table sees identical scenes.
- **AP**: COCO keypoint AP@OKS on all 5,000 val2017 images. It scores whole instances, including false positives and missed joints, which PGA does not see.
- **Native**: the detector's own grouping [2, 4, 6]. **Gap**: native minus the pipeline being scored, so a negative gap means the pipeline is ahead.
- **Autonomous**: no ground truth at inference. **True K**: the number of annotated people with at least one matched detection, used only as a diagnostic.
- **Intervals**: 95 % paired bootstrap over scenes (PGA) or images (AP). They do not cover training noise.

2 Assignment rather than clustering

The MEng work treated grouping as clustering: learn a good embedding, then run k-means. Clustering puts similar points together, and nothing stops it placing two left elbows in one person. A person is better described as a template with 17 typed slots, each holding at most one joint of that type. That is an assignment problem, and the PEng work starts from this reformulation.

Type constraints on clustering. COP-Kmeans [8] adds a cannot-link constraint between joints of the same type. On a standard GAT and COCO ground-truth keypoints, PGA rose from 0.879 for k-means to 0.910. Constrained agglomerative (0.887) and spectral (0.894) clustering also beat plain k-means on COCO.

Optimal transport (SCOT). SCOT defines $K \times 17$ typed slots with learned person prototypes, infinite cost on a type mismatch and Sinkhorn [3] iterations. It reached 0.923 on synthetic data but 0.862 on COCO, below COP-Kmeans. Four variants stayed at 0.83–0.87 and other training strategies

at 0.86–0.88. SCOT-KI replaced the prototypes with k-means centroids of the scene and recovered 6.5 points, reaching 0.926. It stayed 4.2 points below COP-Kmeans on the same embedding, which isolates the Sinkhorn step as the fault. The ranking held on HigherHRNet backbone features too, so the embedding is not the cause.

Type-wise Hungarian assignment (THA). THA replaces Sinkhorn with one optimal matching per joint type. Starting from k-means centroids μ_k, the M_t joints of each type t are matched one-to-one to the K centroids,

$$\pi_t^* = \arg\min_{\pi_t} \sum_{j\,:\,\mathrm{type}(j)=t} \left\| e_j - \mu_{\pi_t(j)} \right\|_2 , \tag{1}$$

by the Hungarian algorithm [5], where e_j is the embedding of joint j. Centroids are then recomputed from the joints of all types, and the steps repeat for up to 10 iterations. Two joints of one type cannot share a person by construction, and the method has no learned parameters. No prior method with this decomposition was found: OpenPose [1] matches pairs of adjacent types instead of one type against person centroids. Table 1 compares these methods on one embedding.

Table 1: Grouping methods on one embedding (COCO-fine-tuned SA-GAT, true K), PGA. Detections are HigherHRNet's.

Method	Synthetic	COCO GT keypoints	Detections
k-means	0.928	0.957	0.942
SCOT, fixed prototypes	0.838	0.861	—
SCOT-KI	—	0.926	—
COP-Kmeans	0.938	0.968	0.955
THA	**0.965**	**0.969**	**0.955**

A feasibility bound on K. A type-constrained grouping with fewer clusters than the most frequent joint type must violate the constraint. The pipeline therefore never lets K fall below that count:

$$K_{\mathrm{lb}} = \max_t M_t, \qquad K = \max(\hat{K}, K_{\mathrm{lb}}), \tag{2}$$

where $\hat{K}$ is the K-head prediction. The rule adds no parameters. Table 2 shows the effect of K rule and grouping method on the unchanged MEng embedding.

Table 2: K rule and grouping method on the MEng embedding, 2,166 HigherHRNet scenes (native 0.9849), PGA.

K rule	Mean K	k-means	COP-Kmeans	THA	Gap (best)
Annotated people (MEng protocol)	2.74	0.9588	0.9634	0.9631	0.0215
K-head alone (MEng autonomous)	2.46	0.9644	0.9657	0.9660	0.0189
K_{lb} alone	2.46	0.9714	0.9753	0.9757	0.0093
Clamp, Equation (2)	**2.55**	**0.9715**	**0.9752**	**0.9755**	**0.0094**
True K (diagnostic)	2.59	0.9750	0.9798	0.9797	0.0052

With no change to the embedding, the autonomous gap fell from 0.0205 to 0.0094 [0.0070, 0.0119]. The K rule contributed about 0.0070 and constrained assignment about 0.0041. In AP, THA with K_{lb} added 0.0805 / 0.1101 / 0.0847 over the MEng pipeline on the three detectors, the largest single step of the work.

THA and COP-Kmeans cannot be separated on real data. Under the clamp the difference is +0.0004 [−0.0004, +0.0011], and both reach 0.955 on detections in Table 1. THA's clear lead is on synthetic data (0.965 against 0.938), and it needs no restarts: 10 restarts gave +0.0000 on synthetic data and −0.0002 on COCO, while COP-Kmeans uses 10. The evidence supports constrained assignment over clustering, not THA over COP-Kmeans on COCO.

3 PG-GAT updates

Table 2 leaves a gap of 0.0052 at true K, so the embedding had to improve as well. That gap is genuine: PG-GAT beats native on the joints HigherHRNet's refine step adds (0.926 against 0.895), and trails it on ordinary heatmap peaks (0.975 against 0.985). Figure 1 shows the resulting pipeline and Table 3 the five changes.

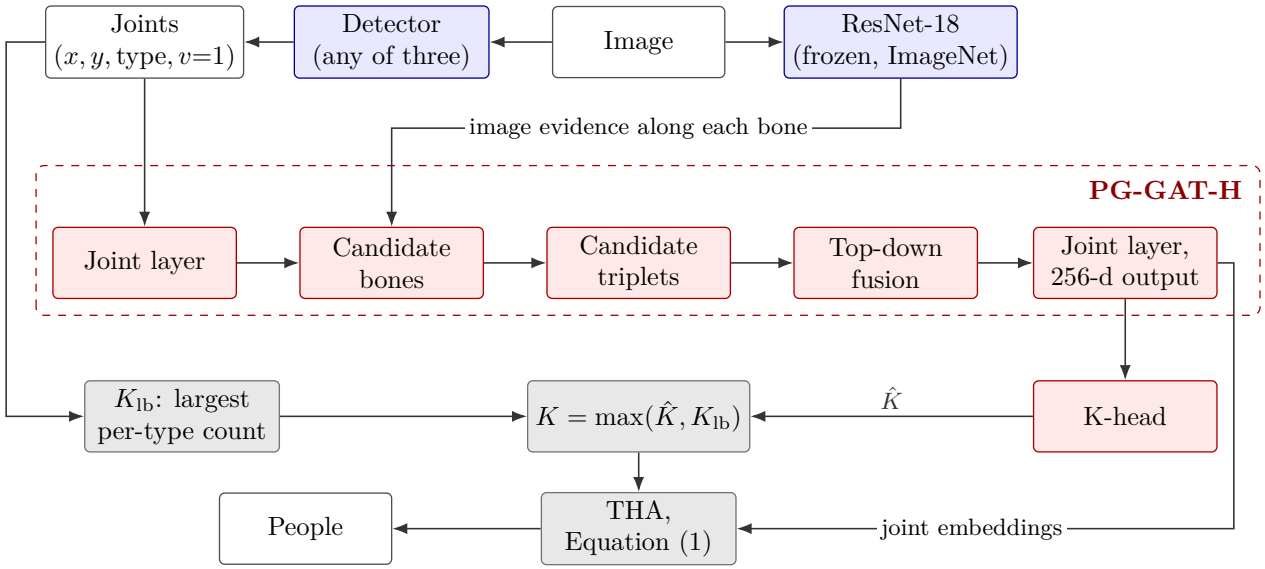


Figure 1: The PEng pipeline. The detector is HigherHRNet, YOLO11x-pose or RTMO-L. Red: trained in this work. Blue: external or frozen. Grey: deterministic, no parameters.

Table 3: The five updates. PGA on HigherHRNet scenes against the previous step unless stated; three values are HigherHRNet / YOLO11x-pose / RTMO-L.

Update	What changed	Effect on detections
Hierarchy (PG-GAT-H)	Joints, candidate bones (3 nearest detections per adjacent type), candidate triplets, then validity-weighted fusion back into joints. No person labels and no voting.	Alone: −0.0017 [−0.0034, +0.0001], despite +0.0098 on COCO GT keypoints. Dropping 20 % of joints removes its advantage.
Detection-like training	Joint dropout $p \sim U(0, 0.4)$ per training graph, visibility fixed at 1.	K_{lb} THA +0.0019 [+0.0004, +0.0034], true K +0.0031 [+0.0013, +0.0049]. Flat control: −0.0004, so the gain is the hierarchy's.
Image evidence	Frozen ImageNet ResNet-18 (stride 16), 5 samples along each candidate bone, zero-initialised branch.	K_{lb} THA +0.0018 [+0.0008, +0.0029], true K +0.0025 [+0.0012, +0.0038]. True-K gain +0.0021 to +0.0026 on all three.
Augmented K-head	K-head on frozen embeddings with the same dropout, selected on held-out COCO train images.	Clamp minus K_{lb} THA: +0.0034 / +0.0038 / +0.0028, intervals above zero. Control on the MEng embedding: −0.0037.
Crowd over-sampling (3×)	Images with 6 or more people go from 12.1 % to 29.2 % of samples, with epoch size unchanged.	Two runs per arm, 6 or more people: +0.0040 / +0.0041 / +0.0030. All scenes: +0.0012 / +0.0009 / +0.0005.

Cost. The grouping pipeline has 3.30M parameters: 479,794 trainable in PG-GAT-H, 35,009 in the K-head and 2,782,784 frozen in the backbone. THA and the clamp add none. These sit on top of the detector, whereas HigherHRNet's own tag head has about 561 parameters [7]. The pipeline is therefore much larger than the grouping it replaces. What it gains is that one model works on any detector without retraining.

4 Results

Table 4 and Figure 2 show the gap after each cumulative stage, one training run per row. The gap falls at every stage, on every detector and in both metrics. The smallest steps are comparable to training noise: stage E overall, and stage D for HigherHRNet AP.

Table 4: Gap to native after each stage. PGA rows B–E use clamp THA; AP rows B–E use K_{lb} THA with every detected joint (threshold 0). HRNet is HigherHRNet.

		PGA gap			AP gap		
	Stage	HRNet	YOLO	RTMO	HRNet	YOLO	RTMO
A	MEng: K-head, k-means	0.0205	0.0210	0.0220	0.1177	0.1360	0.0994
B	+ THA, K bound	0.0094	0.0103	0.0141	0.0371	0.0259	0.0147
C	+ hierarchy, training, K-head	0.0061	0.0063	0.0102	0.0144	0.0102	0.0062
D	+ image evidence	0.0017	0.0034	0.0069	0.0116	0.0056	0.0036
E	**+ crowd over-sampling**	**0.0004**	**0.0017**	**0.0064**	**0.0076**	**0.0050**	**0.0024**

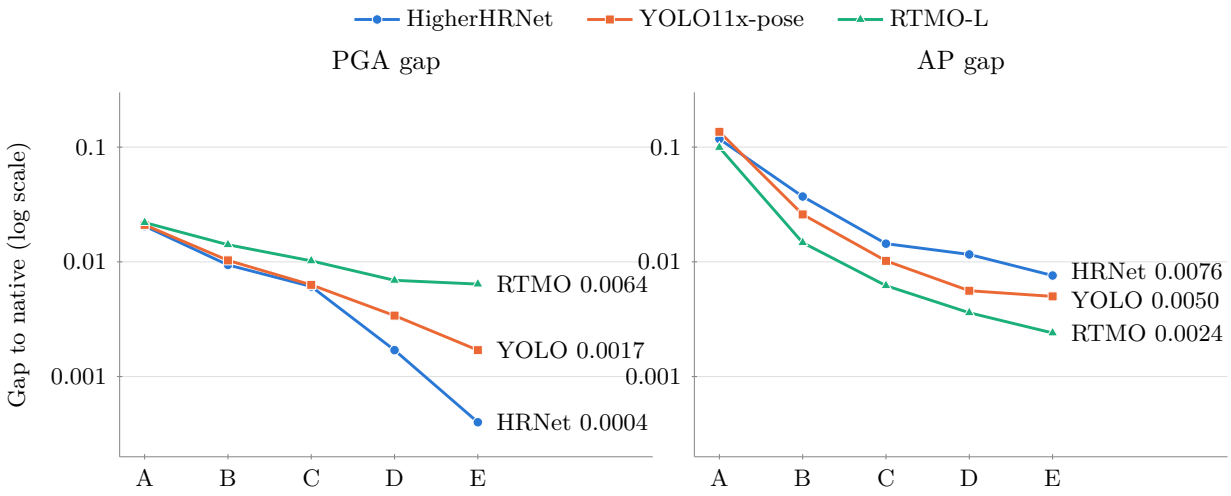


Figure 2: Gap to native after stages A–E of Table 4, on one shared log scale.

The final recipe was trained twice. The first run turned out to be the better one, so its numbers in Table 4 are optimistic by about one noise draw. Tables 5 and 6 give both runs.

Table 5: Final recipe, PGA with THA, two training runs.

	HigherHRNet	YOLO11x-pose	RTMO-L
Native	0.9849	0.9863	0.9961
Autonomous, run 1 / run 2	0.9845 / 0.9840	0.9846 / 0.9837	0.9897 / 0.9895
Autonomous gap, run 1	0.0004 [−0.0016, 0.0025]	0.0017 [−0.0007, 0.0042]	0.0064 [0.0051, 0.0078]
Autonomous gap, run 2	0.0010 [−0.0012, 0.0031]	0.0027 [0.0001, 0.0053]	0.0066 [0.0053, 0.0080]
True-K gap, run 1	−0.0024 [−0.0043, −0.0005]	−0.0005 [−0.0028, 0.0017]	0.0062 [0.0048, 0.0077]
True-K gap, run 2	−0.0005 [−0.0027, 0.0016]	0.0002 [−0.0022, 0.0025]	0.0059 [0.0046, 0.0073]

Table 6: Final recipe, AP with K_{lb} THA at threshold 0, two training runs. Intervals are BCa. On RTMO-L run 2 the BCa interval is unstable; its percentile interval is [0.0004, 0.0040].

	HigherHRNet	YOLO11x-pose	RTMO-L
Native	0.5834	0.6666	0.6460
Pipeline, run 1 / run 2	0.5758 / 0.5740	0.6616 / 0.6613	0.6436 / 0.6426
Gap, run 1	0.0076 [0.0035, 0.0100]	0.0050 [0.0027, 0.0082]	0.0024 [0.0018, 0.0049]
Gap, run 2	0.0094 [0.0064, 0.0130]	0.0053 [0.0030, 0.0083]	0.0034 [0.0030, 0.0068]
K-head clamp row, gap, run 1	0.0182 [0.0132, 0.0208]	0.0225 [0.0185, 0.0249]	0.0208 [0.0176, 0.0251]
AP with perfect grouping (estimate)	0.627	0.678	0.646

Where the AP gap comes from. It is single wrong joints in people who are found, mostly in crowds, and not missed people or ranking. At OKS 0.50 the pipeline is level with or ahead of native on all three detectors. Giving its instances the native scores recovers nothing. It loses 528 / 164 / 50 correct joint slots and wins 181 / 27 / 1. Of the people who fall below OKS 0.75, about 70 % are in scenes with 6 or more people, and about 40–45 % differ from native in one joint. In about 70 % of lost slots the joint native kept is the more confident one, and the model never sees confidence.

What the evidence supports.

- Constrained assignment and the K bound improve on clustering by a wide margin on the same embedding.
- PGA parity with HigherHRNet holds in both runs. YOLO11x-pose is within 0.002–0.003: its interval includes zero in one run of two, so parity is not claimed.
- RTMO-L keeps a real gap of about 0.006 PGA even at true K.
- The AP gap is above zero on all three detectors in both runs, under both interval methods.
- Crowd over-sampling is a replicated gain of 0.003–0.004 PGA in scenes with 6 or more people, and at most 0.0013 overall. Doubling it to 6× adds nothing measurable.

Limits of these claims.

- **Training noise.** Two runs of one recipe differ by about 0.001 PGA overall. On HigherHRNet AP they differ by up to 0.004: four runs of 3× and 6× give gaps of 0.0055–0.0096. Scene and image intervals do not cover this, so a gap should be quoted as a range over runs.

- **Being ahead at true K.** The lead over HigherHRNet holds for run 1 only and is not claimed.
- **K rule differs by metric.** The AP row uses K_{lb}, which at threshold 0 equals the detector's own instance count. It uses no ground truth, but it borrows the detector's person count and re-shuffles the detector's joints. Adding the K-head (the PGA rule) raises the AP gap to about 0.02, because the head over-counts on the false-positive-heavy pool.
- **No untouched test set.** All numbers are on val2017, the same images that guided the design decisions.
- **"No retraining" is about training only.** No model was trained on any detector's output, but the YOLO11x-pose and RTMO-L rows did guide later choices, such as crowd over-sampling.
- **Early numbers.** The early assignment results (Table 1) stand as recorded at the time and were not part of the verification. The lab book also holds two sets of SCOT variant numbers; the range quoted covers both, and the conclusion is the same under either.
- **Runtime.** Inference time of the pipeline has not been measured.
- **Single runs remain.** The hierarchy's attribution rests on one run each of PG-GAT-H + training and the flat control. The K-head of record was the best of several, so its K-head AP rows are optimistic.

5 All experiments

Table 7 lists the PEng experiments by phase, as recorded in the lab book. Three values are HigherHRNet / YOLO11x-pose / RTMO-L, and Δ is the new arm minus the baseline. "Autonomous" is clamp THA.

Table 7: Every PEng experiment, its result and the decision taken.

Experiment	Result	Decision
Assignment methods (COCO GT keypoints, true K)		
Type constraints on clustering	k-means 0.879; COP-Kmeans 0.910, spectral 0.894, agglomerative 0.887	Kept
SCOT (typed optimal transport)	Synthetic 0.923, COCO 0.862	Not adopted
SCOT variants: residual, adaptive, unbalanced, dustbin	COCO 0.83–0.87, none above COP-Kmeans	Not adopted
SCOT training: joint COCO fine-tune, frozen GAT, standard GAT	COCO 0.884, 0.861, 0.881	Not adopted
SCOT-KI (k-means prototypes)	0.926, against COP-Kmeans 0.968 on one embedding: Sinkhorn is the fault	Not adopted
Methods on HigherHRNet backbone features	k-means 0.789, COP-Kmeans 0.891, THA 0.884, SCOT-KI 0.825: same ranking, so not the embedding	Diagnostic
THA	Synthetic 0.965, COCO 0.969, detections 0.955	Adopted
K-head training	Frozen on COCO: exact K 79.6 %, COP-Kmeans 0.950. Joint: 55.5 %, 0.677, though a later re-run lost only 0.015 PGA	Frozen kept
Autonomous SCOT pipeline	Detections, K-head: SCOT 0.800, COP-Kmeans 0.816	Not adopted
Test-time compute (MEng embedding)		
THA with 10 restarts	+0.0000 synthetic, −0.0002 COCO GT	Not adopted
K selection by silhouette	K exact 39–46 %, costs about 0.15 PGA	Not adopted

Experiment	Result	Decision
K feasibility clamp	Detections 0.9644 to 0.9755	Adopted
Pipeline		
Fixed-scene gap table	Gap 0.0094 [0.0070, 0.0119]: K 0.0042, embedding 0.0052; K_{lb} alone ties the clamp	Benchmark
HigherHRNet without refine	Same peaks: PG-GAT 0.9745, tag matching 0.8666; AP falls 0.492 to 0.386, so refine also detects. Final pipeline on these peaks: 0.9823	Diagnostic
PG-GAT-H, first run	COCO GT THA +0.0098, detections −0.0017	Kept
Shift diagnostic	Dropping 20 % of joints removes the advantage; jitter costs nothing	Diagnostic
Detection-like training with flat control	K_{lb} THA +0.0019 with hierarchy, −0.0004 flat	Adopted
Gap by joint origin	Deficit is on heatmap peaks (0.985 against 0.975), so the missing information is in the image	Diagnostic
K from bone validity	THA 0.8465 to 0.9722, against K_{lb} 0.9755; pieces 1.6–9 % impure	Not adopted
Coordinate frame check	Letterbox frame −0.0007 [−0.0015, +0.0001]	Immaterial
Image evidence along bones	K_{lb} THA +0.0018, true K +0.0025	Adopted
Augmented K-head	Clamp minus K_{lb} THA +0.0034 / +0.0038 / +0.0028	Adopted
Detector swap (no retraining)	K_{lb} THA over stage B +0.0041 / +0.0031 / +0.0044	Evaluation
AP@OKS on three detectors	Threshold 0 with K_{lb} THA is the best AP rule	Evaluation
Crowd over-sampling, 3×	Autonomous +0.0013 / +0.0017 / +0.0005, one run	Adopted
Verification and noise		
Verification of every number	2,282 metrics on the CPU; largest change 0.0007 PGA, 0.0024 AP; no conclusion changes	CPU values canonical
Diagnostics	With K right, error 0.79 % / 0.74 % / 0.48 % against native 1.67 % / 0.99 % / 0.06 %; weak on large people and in crowds	Diagnostic
AP confidence intervals	AP gap above zero on all three; BCa and percentile both required	Method kept
Fine-tune seed repeat	Noise about 0.001 PGA; 3 of 12 same-recipe intervals exclude zero	Two runs per arm
No-crowd repeat (2×2 runs)	6+ people +0.0040 / +0.0041 / +0.0030; AP in 6+ images +0.0051 / +0.0035 / +0.0007	3× kept
Improvement attempts on the final pipeline		
K-head with per-type count histogram	Autonomous −0.0006 / −0.0011 / −0.0009, despite better held-out K (0.912 against 0.899)	Not adopted
Same-type edges in the joint graph	Autonomous −0.0009 / −0.0001 / +0.0006; true K, 6+ −0.0025 / +0.0032 / +0.0034; not separable from noise	Not adopted
Scale augmentation	Autonomous −0.0002 / −0.0001 / +0.0000; target slice unmoved	Not adopted
Joint-level image features	Autonomous −0.0021 / −0.0018 / −0.0014, all intervals below zero; target joints worse	Not adopted
AP-gap anatomy	Single wrong joints in crowds, not ranking; model never sees confidence	Diagnostic

Experiment	Result	Decision
Confidence-aware THA	K_{lb} AP +0.0007 / +0.0004 / −0.0010, none robust; clamp AP +0.006 to +0.016, still below K_{lb}; PGA change at most 0.0004	Not adopted
Crowd over-sampling, 6× (2×2 runs)	AP K_{lb} mean Δ +0.0009 / +0.0004 / −0.0005, below run-to-run differences; no PGA row passes	Not adopted

References

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